Proposed Contribution for the Plate Detection and Reading Models

CityPulse: Intelligent Parking & Traffic Control Platform

CSE499B, Section 15, Group 03

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Two models read a plate today

Camera image

Parking or traffic camera frame

→

YOLO26n detector

Finds the plate

→

YOLO26s reader

Finds each character on the crop (103 classes)

→

Plate number

Joined into Bangla and English text

0.981

detector mAP@0.5

75.7%

plates read fully correct

4.4%

character error rate

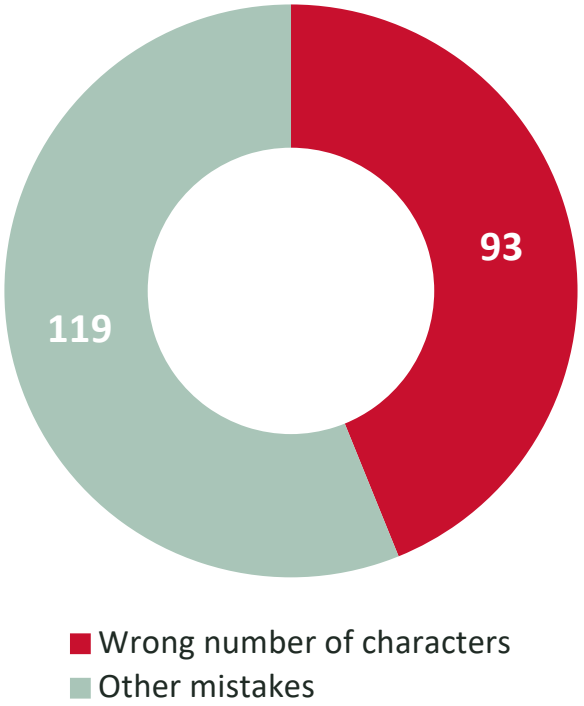
5.5 ms

both models, Tesla T4

We also reproduced ViT + BanglaBERT (Hasin et al., 2026) on the same data: 31.2% plates correct, 21.2% CER, 43.7 ms. It often got the district right but the digits wrong, so the YOLO26s reader stays our baseline. All readers are scored on the same 218 test plates.

Most mistakes are about the plate's structure

212 plates read wrongly



1

Wrong length

93 of 212 errors. Seven digits instead of six (the dash read as a one), a letter detected twice, or no letter at all. A real plate cannot look like this, but the model does not know.

2

Dark plates

A dim basement car park in Dhaka gave “Dhaka Metro 21-8842”: the series letter is missing completely.

3

New image sources

A detector trained like the earlier paper: 0.991 mAP on its own test set, 0.910 on our merged one.

4

Rare classes

Some districts and letters appear in only a few training crops.

What is already published (22 papers read)

Idea	Papers	Reported result
One network for detection and reading	Li et al. 2017; Xu et al. 2018	96.1% end to end (Chinese plates)
Cascaded YOLO models for Bangla plates	Sarker and Alam 2026; Bristy et al. 2025	0.949 character mAP; 94.3% full plates
Low-light enhancement trained with the detector	Xiong et al. 2026; Guo et al. 2020	mAP@0.5:0.95 from 54.1 to 61.3
Straightening tilted plates before reading	Silva and Jung 2018	84.6% to 98.4% on oblique plates
Supervising each character's position	Zhang et al. 2021	83.6% to 97.7% on AOLP
Learning the plate format with a language model	Shabaninia et al. 2025	97.1% on Iranian plates
Fixed layout rules after the character detector	Laroca et al. 2021	96.9% over eight datasets
Removing train/test duplicates	Laroca et al. 2023	Error rate more than doubles on fair splits

None of these can be claimed as new by us.

What we did not find for Bangla plates

The Bangladeshi papers either detect characters and sort them left to right, or run an OCR engine on the plate crop. None of them:

1. **Builds the plate layout into the model**

   Two rows, district and letter on top, six digits below

2. **Handles low light or tilt inside the model**

   Not as a separate pre-processing step

3. **Removes duplicate images between train and test**

   So reported results may be too high

4. **Reports results across datasets and conditions**

   Dark, tilted, small, other sources

Idea 1: read a plate by redrawing it

BRTA fixes the plate design (white background, one font, district and letter on top, six digits below), so any plate number can be drawn in software.

1 Reader proposes

Take the reader's top few readings

2 Draw each one

A small differentiable renderer draws that plate

3 Fit to the image

Adjust 4 corners, brightness and blur with a few gradient steps

4 Keep the best fit

The reading whose drawing matches the photo wins

- **One step for tilt, low light and blur:** instead of three separate modules
- **A reliability score for free:** a large leftover difference means the reading is doubtful
- **Synthetic data for rare classes:** the same renderer draws plates for rare districts and letters

What is new

Rendered plates have only been used to make training data (PixTextGAN 2019, diffusion 2025). We found no work that reads a plate by fitting a drawn plate to the image.

Idea 2: use the vehicle type to choose the letter

On a Bangladeshi plate the series letter depends on the vehicle (BRTA registration rules):

Motorcycle	A, Ha, La (by engine size)
Private car	Ka, Kha, Ga, Bha
CNG auto-rickshaw	Tha
Bus	Ba

Plate colour shows the use: white for private, green for commercial.

How it works

- Add a head to the detector that predicts vehicle type and plate colour.
- Limit the reader's letter to the ones allowed for that vehicle.
- A motorcycle can never be read with a car letter.

Small extension: many frames

In traffic a car stays visible for many frames, so the character scores of one tracked plate can be combined. The Bangla work we read uses single images only.

How we will test it

Variant	What it tells us
Current YOLO26n + YOLO26s	Baseline (75.7% correct, 4.4% CER)
+ fixed plate rules	How much simple rules already help
+ redrawing	Effect on dark, tilted, blurred plates
+ vehicle and colour head	Effect on letter errors
Both together	Final model

Same data, same splits, one change at a time.

Metrics: full-plate accuracy, character error rate, detection mAP@0.5, time per plate.

Reported separately

- **Dark images**
  from image brightness
- **Tilted plates**
  from the slope of the character rows
- **Small plates**
  from the plate size
- **Other dataset**
  train on one, test on the other
- **Our Dhaka camera**
  real CityPulse parking images

Order of work and risks

1

Vehicle and colour head

The simpler part, done first

2

Renderer for re-ranking

Re-rank the reader's top readings

3

Renderer for synthetic data

Plates for rare districts and letters

4

Into CityPulse

Replace the model in the AI service

Risks

Real plates are dirty or faded
The drawing never matches exactly. If raw pixels fail, compare edge maps.

Letter table not yet verified
It comes from public sources; must be checked against BRTA rules.

Mostly daylight data
A low-light gain may only show on the dark subset.

22 papers, not the whole field
We say novel “to the best of our knowledge”.

We would like your advice

1. Is this a suitable direction for the AI contribution of the project?
2. Should we concentrate on one of the two parts: redrawing, or the vehicle-aware letter?

References

[1] N. Hasin et al., Bangla license plate recognition using YOLO and vision-language OCR, arXiv:2603.10267, 2026.

[2] H. Li, P. Wang, C. Shen, Towards end-to-end car license plates detection and recognition, arXiv:1709.08828, 2017.

[3] Z. Xu et al., Towards end-to-end license plate detection and recognition, ECCV 2018.

[4] P. Sarker, S. M. S. Alam, Cascaded YOLO-based approach for Bangladeshi license plates, PLOS ONE 21(8), 2026.

[5] N. J. Bristy et al., An optimized deep learning approach for ALPR, IDAA 2025.

[6] W. Xiong et al., CSCM-YOLOv8 and CSM-LPRNet in complex scenarios, PLOS ONE 21(1), 2026.

[7] C. Guo et al., Zero-reference deep curve estimation for low-light enhancement, CVPR 2020.

[8] S. M. Silva, C. R. Jung, License plate detection and recognition in unconstrained scenarios, ECCV 2018.

[9] Y. Zhang, Z. Wang, J. Zhuang, Efficient LPR via holistic position attention, AAAI 2021.

[10] E. Shabaninia et al., Layout-independent LPR via integrated vision and language models, arXiv:2510.10533, 2025.

[11] R. Laroca et al., Efficient and layout-independent ALPR based on YOLO, IET ITS 15(4), 2021.

[12] R. Laroca et al., Do we train on test data? Near-duplicates in LPR, IJCNN 2023.

[13] C. Wu et al., PixTextGAN: structure aware text image synthesis for LPR, IET Image Processing, 2019.

[14] License plate images generation with diffusion models, arXiv:2501.03374, 2025.

[15] Vehicle registration plates of Bangladesh, Wikipedia.